

1(a)(i)	<i>any one from:</i> <i>glucose</i> is activated / made reactive ; becomes unstable ; cannot leave cell ; energy level increased ; concentration gradient maintained ;	1
1(a)(ii)	<i>any one from:</i> carrier / acceptor / provider, of, hydrogen (atoms) / electrons or provider of H ions and electrons ; R H ions alone used for redox reactions ; used to reduce, ethanal / pyruvate ;	1
1(a)(iii)	substrate-linked / substrate level ;	1

Question	Answer	Marks
1(b)	any three from: 1 (inner) mitochondrial membrane impermeable (to pyruvate) ; 2 carrier (protein) ; A symport / MPC (mitochondrial pyruvate carrier) 3 lower concentration of pyruvate outside mitochondrion ora or against concentration gradient ; 4 oxygen available ; 5 ATP / energy ;	3
1(c)	any three from: 1 decarboxylation ; A description 2 dehydrogenation / oxidation, (of pyruvate) ; A description 3 reduced NAD produced ; 4 formation of, acetyl coenzyme A / acetyl CoA ;	3